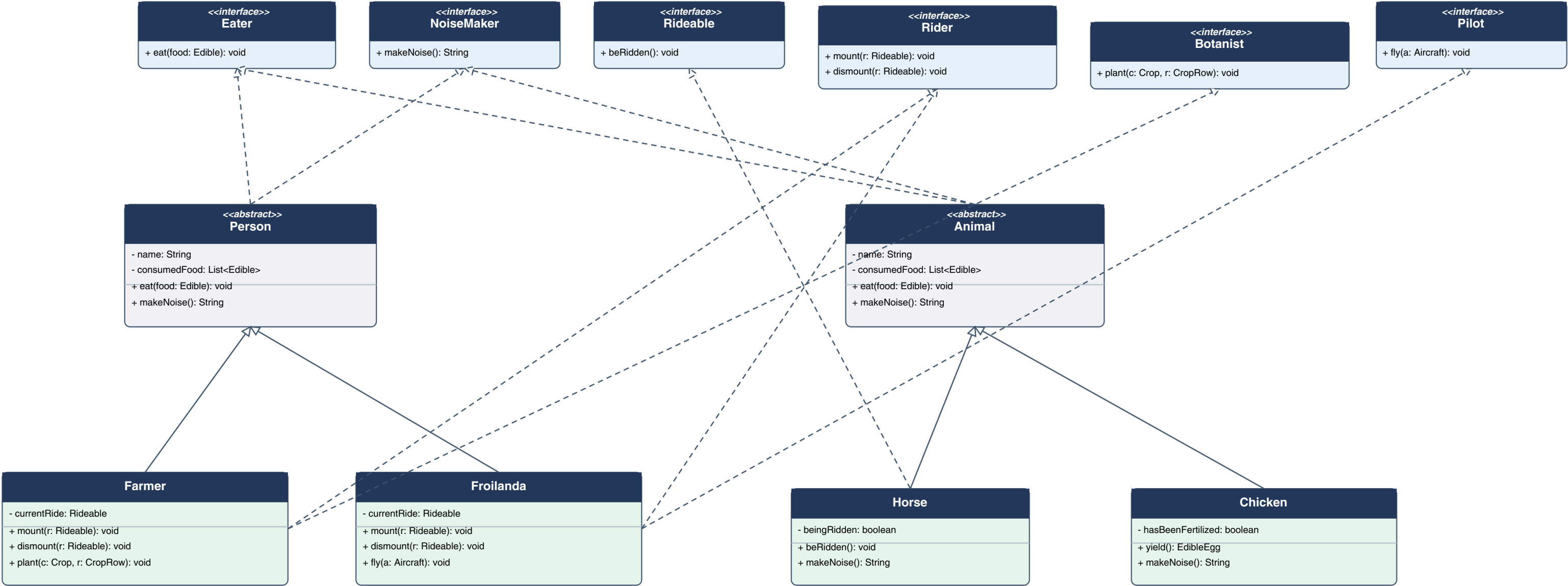


1. Core Abilities, People, and Animals

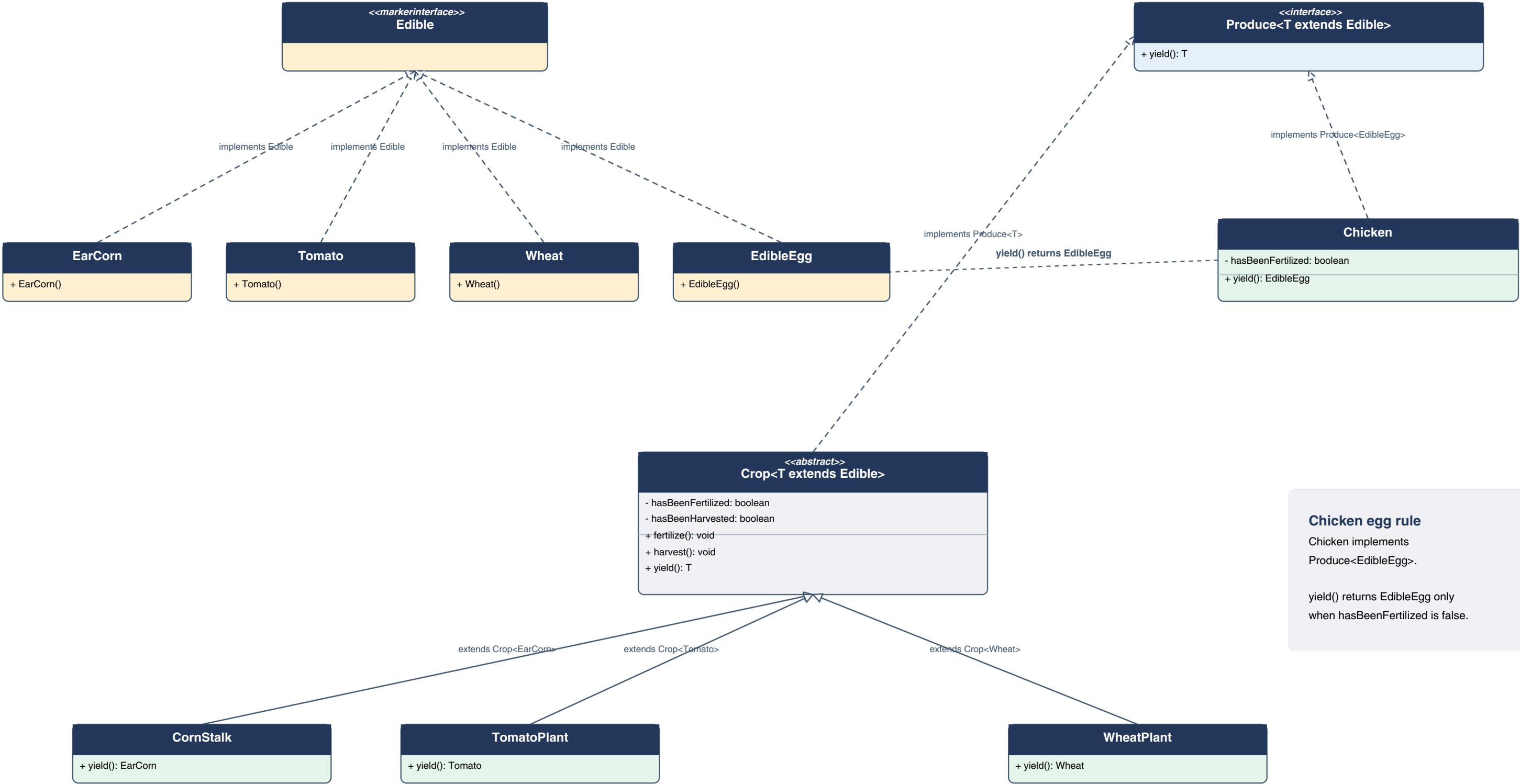
Dashed hollow arrow = implements | Solid hollow arrow = extends



Chicken also implements Produce<EdibleEgg> (shown on Page 2).

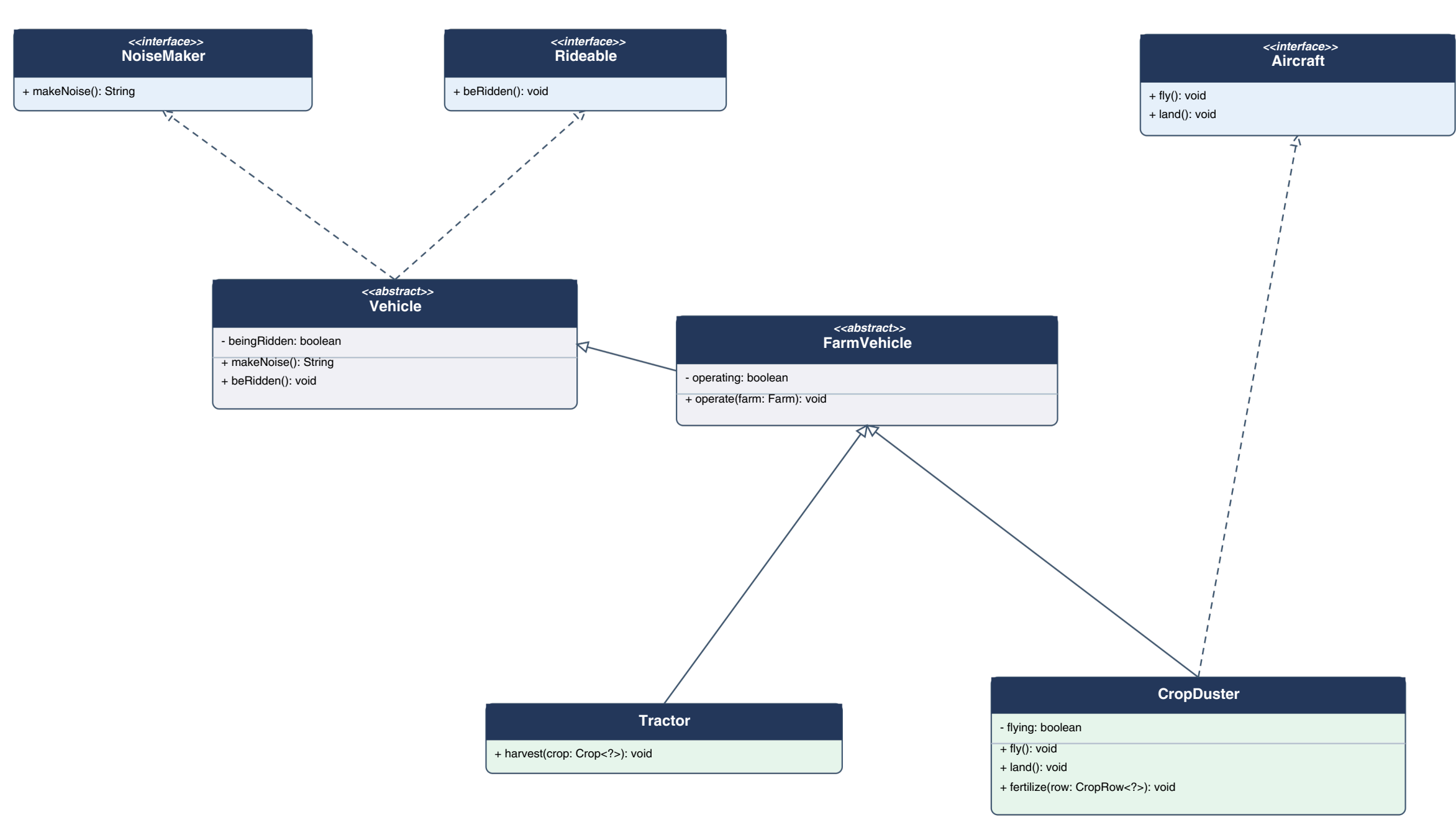
2. Edible Types, Produce, Crops, and Generics

Edible is a marker interface; dashed arrows show exactly which food classes implement it



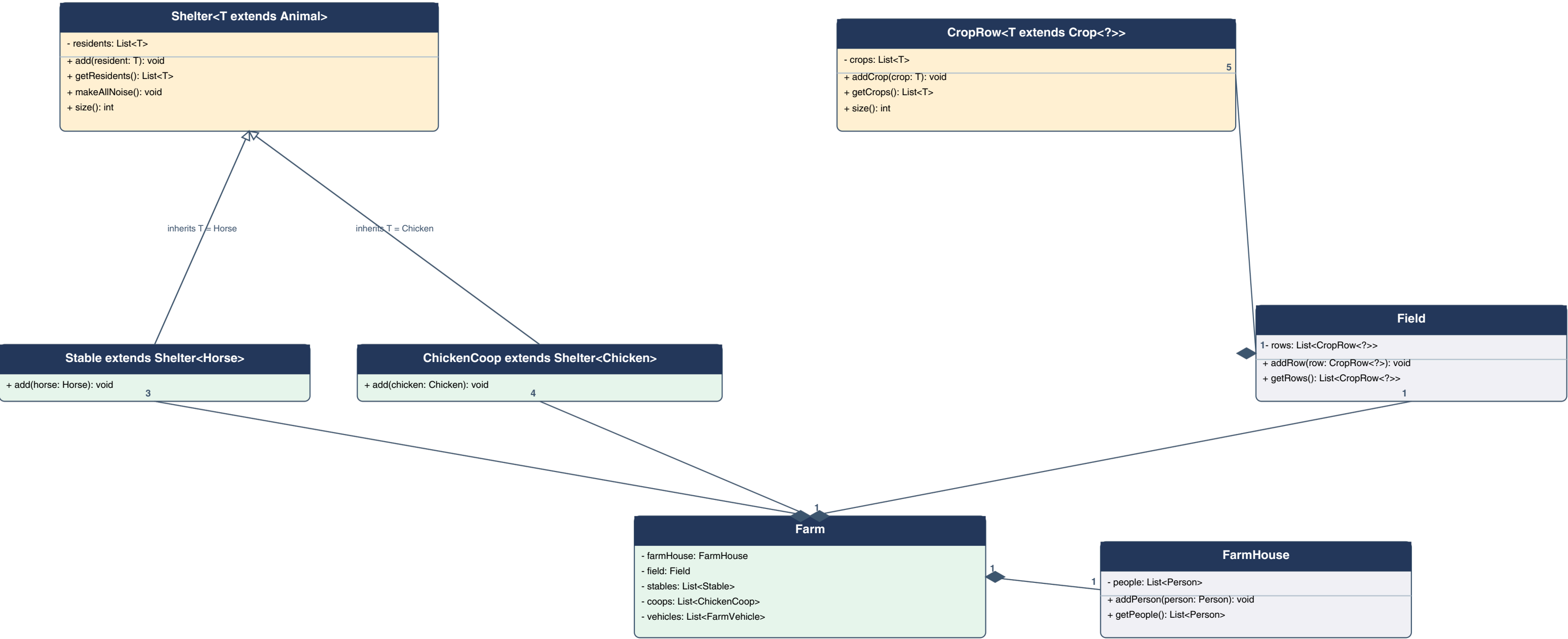
3. Vehicle Hierarchy and Multiple Inheritance

CropDuster extends one class and implements Aircraft to avoid Java's multiple-class inheritance restriction



4. Bounded Generic Containers and Farm Composition

The type argument is a compile-time rule: Shelter<Chicken> accepts Chicken, never Horse



Compile-time guarantees

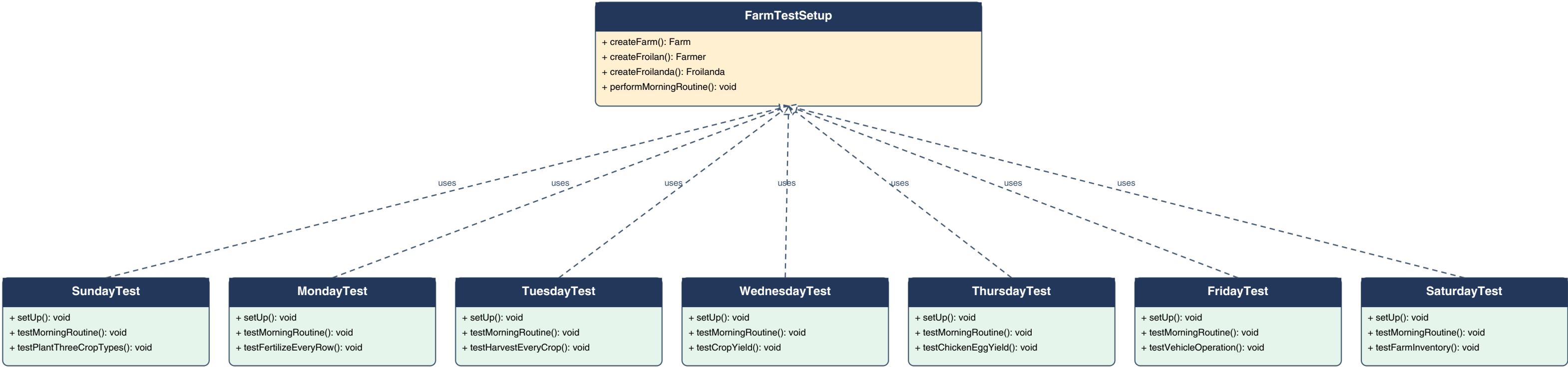
Stable is Shelter<Horse>: `add(new Horse())` compiles; `add(new Chicken())` does not compile.

ChickenCoop is Shelter<Chicken>: `add(new Chicken())` compiles; `add(new Horse())` does not compile.

CropRow<CornStalk> accepts CornStalk and returns `List<CornStalk>` without casting. Raw types are not used.

5. Weekly Test Structure

AseparateJUnittestclassisrequiredforeachday;alldaysreuse one farm setup



Daily behavior to verify

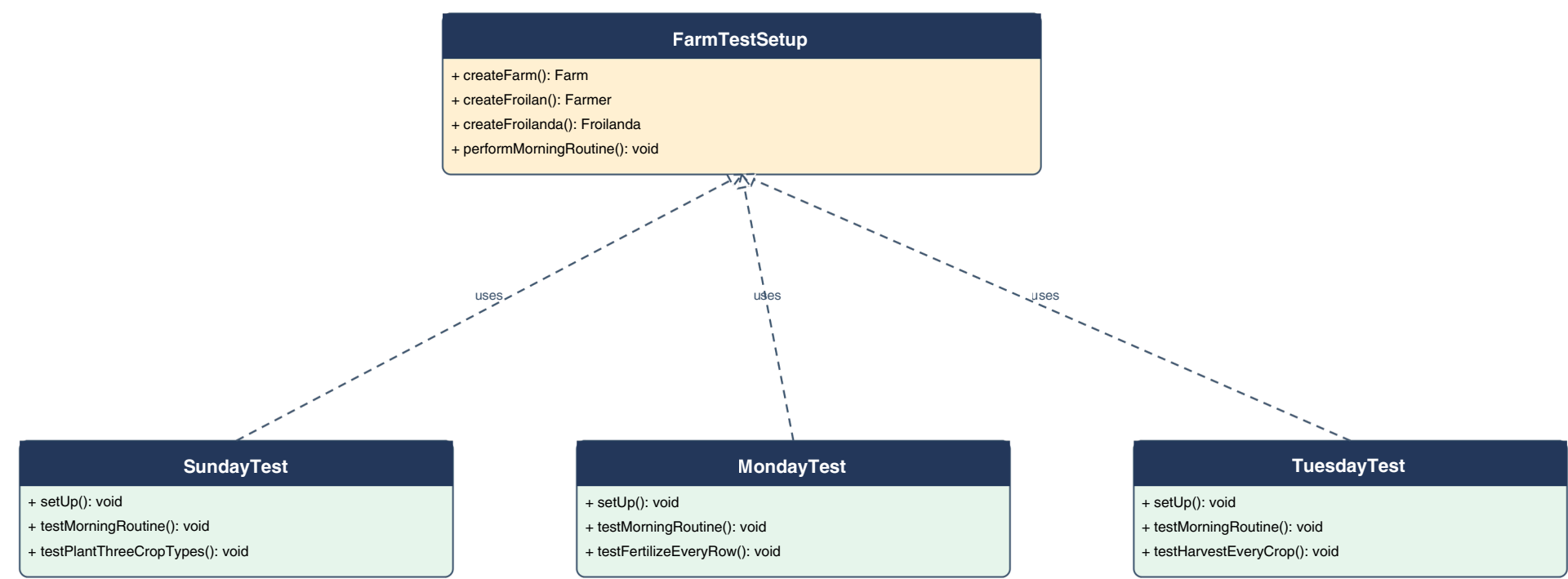
Every morning: Froilan and Froilanda ride every horse; each of 10 horses eats 3 EarCorn; Froilan eats 1 EarCorn, 2 Tomato, 5 EdibleEgg; Froilanda eats 2 EarCorn, 1 Tomato, 2 EdibleEgg.

Sunday: plant 3 different crop types in rows 1, 2, and 3. Monday: CropDuster flies and fertilizes every CropRow. Tuesday: Tractor harvests every Crop in every CropRow.

Sunday-Saturday: a separate test class is required for every day. Sunday-Tuesday use the provided activities; Wednesday-Saturday complete the team's typical work-week simulation.

All tests together must achieve at least 80% coverage and assert state changes, not merely call methods.

5. Sunday Through Tuesday Test Structure



Daily behavior to verify

Every morning: Froilan and Froilanda ride every horse; each of 10 horses eats 3 EarCorn; Froilan eats 1 EarCorn, 2 Tomato, 5 EdibleEgg; Froilanda eats 2 EarCorn, 1 Tomato, 2 EdibleEgg.

Sunday: plant 3 different crop types in rows 1, 2, and 3. Monday: CropDuster flies and fertilizes every CropRow. Tuesday: Tractor harvests every Crop in every CropRow.

Only Sunday through Tuesday are included on this UML page; no Wednesday through Saturday test classes are shown.

The tests should assert state changes, not merely call methods.